Linrui Ma

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Education

- **Massachusetts Institute of Technology (MIT), Cambridge, MA** *Aug 2025 – Present*
  B.E. in Artificial Intelligence & Decision Making & B.S. in Chemistry (Double Major).
  – GPA: 5.0/5.0
  – Core Courses: Deep Learning; Multimodal AI; Symmetry and its Application to Machine Learning; Representation, Reasoning, and Inference in AI; Algorithms; AI & Decision Making in Medicine.
- **Tsinghua University, Beijing, China** *Aug 2024 – Jun 2025*
  Undergraduate student in Yao class, majoring in Computer Science and Artificial Intelligence.
  – GPA: 3.87/4.00
  – Core Courses: Linear Algebra; Programming in C/C++; Artificial Intelligence: Principles & Techniques; Mathematics for CS & AI; Computer Systems; Introduction to LLM Applications.
- Skills: C/C++, Python, TypeScript, PyTorch, NumPy, Matplotlib, LaTeX, Git, CMake, JAX, Linux.

Research and Experiences

- **Software Development Engineer Intern, AWS Transform, Amazon (Boston, MA)** *Jun 2026 – Present*
  Building model-comparison infrastructure for the LLM-based orchestrator behind AWS Transform's VMware migration workflows. Designing automated evaluation scenarios and CloudWatch metrics across latency/cost, tool-calling, and reliability so model selection can be data-driven; implementing the pipeline in Python and TypeScript (feature owned end-to-end under SDE I mentorship).
- **Undergraduate Researcher, MIT CSAIL (Cambridge, MA)** *Feb 2026 – Present*
  Advisor: Prof. Kaiming He
  Working on pixel-space generative models in the MiniT2I line, including adapting the framework toward image editing and contributing LoRA-based downstream adaptation. Built a large-scale image-editing dataset pipeline to Google Cloud Storage; diagnosed OOM failures on ∼300GB tarballs and redesigned the flow with streaming shard extraction for reliable full-dataset upload.
- **Undergraduate Researcher, MIT CSAIL (Cambridge, MA)** *Sep 2025 – Present*
  Advisor: Prof. Regina Barzilay
  Collaborate with Ph.D. candidates under Prof. Regina Barzilay's supervision to design deep learning models that predict cell state changes in response to various cellular perturbations. Developing probabilistic models for selective protein binder design.

Publications

- **ProtQueSt: Query-Conditioned Retrieval-Augmented Generation for Protein Function Annotation** *ICML 2026*
  Linrui Ma*, Yiwei Liang*, Yishu Yu**, Chuhan Joyce Qi**
  ICML 2026 GenBio Workshop (Spotlight); ICML 2026 FM4LS Workshop (Poster).
  Link to paper: `https://openreview.net/forum?id=zUqknDsV1t`.
  First-authored RAG framework that pairs structure-aware retrieval with a query-conditioned contrastive retriever (FiLM + query-pooled negatives), reframing protein function annotation as query-dependent sequence understanding; Entity-BLEU 48.79 (+37% vs RAPM) on Prot-Inst-OOD.
- **MiniT2I: A Minimalist Baseline for Text-to-Image Generation** *Technical Blog, 2026*
  Xianbang Wang†, Hanhong Zhao‡, Yiyang Lu‡, Kangyang Zhou, Linrui Ma, and Kaiming He§
  (†project lead; ‡core contributor; §supervisor)
  Link to blog: `https://peppaking8.github.io/#/post/minit2i`.
  Academic-scale pixel-space T2I baseline (MM-JiT). Contributed the LoRA downstream-adaptation extension—fine-tuning on small style datasets (e.g., Naruto/Pokémon) and producing the adaptation artifacts—and contributed code/notebooks to the public PyTorch release.

Projects

- **NeuroMusic: EEG-Driven Therapeutic Music Generation** *Feb 2026 – May 2026*
  Link to project: `https://github.com/JerryLin828/neuromusic`.
  Built an EEG-to-music pipeline that predicts valence/arousal (TSCeption), applies affective pivoting toward therapeutic targets, and synthesizes music with MusicGen/MusicLDM. Collaborated with 2 peers.
- **Representation Efficiency in Neural Networks** *Oct 2025 – Dec 2025*
  Link to project: `https://github.com/bowenyu066/language-shapes-reasoning`.
  Studied how representation choice affects Transformer reasoning efficiency on multilingual math benchmarks; showed 5–10% token efficiency gains and attributed bottlenecks to training distribution via LoRA. Collaborated with 2 peers.
- **Labotex: An Intelligent Agent for Lab Reports** *Mar 2025 – Jun 2025*
  Link to project: `https://github.com/Kehan-Liu/LABOTEX`.
  Built an AI agent that turns experiment PDFs and CSV data into structured LaTeX lab reports using VLMs and chat models, with a Web UI for one-click PDF/LaTeX export. Collaborated with 2 peers.
- **High-Performance Matrix Multiplication with Multithreading & Distributed RPC** *Mar 2025 – Jun 2025*
  Implemented a high-performance matrix multiplication system in C with cache blocking, SIMD, multithreading, and cross-machine RPC; benchmarked for accuracy and speed. Collaborated with 2 peers.

Key Awards

- **1st Place, MIT Informatics Tournament (MITIT) 2025 Winter Contest** (Beginner's Round) *Dec 2025*
- **Tsinghua University Leading Talents Scholarship** (Top 1%, First Class Scholarship for Freshmen) *Sep 2024*
- **56th International Chemistry Olympiad (IChO) Gold Medalist** (*2nd in theory. Captain of Chinese Team*) *Jul 2024*
- **37th Chinese Chemistry Olympiad (CChO) Gold Medalist** (*2nd place nationally*) *Nov 2023*